

Mission Data Dashboard

Create the clearest and most useful dashboard for a rover test data set.

Advanced	2 class periods	Competition
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Engineering Context

A mission manager needs to make a quick go/no-go decision using data from speed, temperature, battery voltage, sensor faults, and task completion.

What You Will Practice

- Select meaningful data displays
- Avoid misleading graphs
- Create performance thresholds
- Communicate an engineering decision

What You Need

- Instructor-provided rover test data
- Spreadsheet or visualization software
- Mission requirements
- Engineering notebook

Student Instructions

- 1 Review the data dictionary and mission success criteria.
- 2 Identify the five most decision-relevant measures.
- 3 Choose an appropriate chart, table, or status indicator for each.
- 4 Create threshold colors or flags based on stated requirements.
- 5 Build a one-page dashboard.
- 6 Write a go/no-go recommendation using evidence from the dashboard.
- 7 Exchange dashboards and have another team answer three mission questions.
- 8 Revise any display that caused confusion.

Engineering Requirements

- One-page dashboard
- At least three different visualization types
- Every visual must include labels and units
- Thresholds must come from mission criteria
- No decorative chart may obscure data
- Recommendation must be traceable to the dashboard

Constraints & Safety

- Do not alter source data
- Clearly identify missing data
- Avoid color-only communication when possible

What You Submit

- Dashboard
- Data-selection rationale
- Go/no-go recommendation
- Peer usability results
- Revised final version

Reflection Questions

- 1 Which visual supported the fastest decision?
- 2 What information was unnecessary?
- 3 How did you display uncertainty or missing data?
- 4 What graph choice could mislead a viewer?

Engineering Tip Design the dashboard around decisions the user must make, not around every variable available.	Common Mistake Using a visually impressive but inappropriate graph can hide patterns or exaggerate differences.
Stretch Goal Create a live version that automatically updates when new rows are added.	Career Connection Flight-test, systems, and mission-operations teams use dashboards to monitor vehicle health and test readiness.